

Python Poetry

- Poetry is a tool for **dependency management and packaging** in Python.
- It helps you create, manage, and publish Python projects with ease.

Advantages:

- Simple pyproject.toml file replaces requirements.txt, setup.py, etc
- Keeps track of packages **added or removed** along with their **exact versions**
- Easy Installation
- Automatically creates isolated environments
- Reproducible builds across machines and environments due to **poetry.lock** file
- Allows package and publish to **PyPI**

```
my_project/
|
├── pyproject.toml #Project metadata +
dependencies
├── poetry.lock    # Exact versions (like
package-lock.json)
```

Commands:

```
pip install poetry    → Poetry package installation
# To create a new project
poetry new my_project
# To use poetry in existing project
poetry init
# Add dependency packages
poetry add pandas
# Remove installed packages
poetry remove pandas
# To install all packages dependency from poetry file
poetry install
# Activate Shell
poetry shell
# Execute files inside virtual environment
poetry run python yoursript.py
# Publish to PyPI
poetry publish --build
```

Python UV

- UV is a tool for **dependency management and packaging** in Python.
- It helps you create, manage, and publish Python projects with ease.

Advantages:

- Simple pyproject.toml file replaces requirements.txt, setup.py, etc
- Keeps track of packages **added or removed** along with their **exact versions**
- Easy Installation
- Automatically creates isolated environments
- Reproducible builds across machines and environments due to **uv.lock** file
- Allows package and publish to **PyPI**

```
my_project/  
├── pyproject.toml #Project metadata +  
dependencies  
├── uv.lock      # Exact versions (like  
package-lock.json)
```

Commands:

```
pip install uv    → uv package installation  
# To create a new project  
uv init my_project  
# To use poetry in existing project  
uv init --bare  
# Add dependency packages  
uv add pandas  
# Remove installed packages  
uv remove pandas  
# To install all packages dependency from uv file  
uv sync  
# Execute files inside virtual environment  
uv run python yoursript.py  
# Publish to PyPI  
uv build  
uv publish
```